3.5 Diode Applications & Circuits

3.5.1 Temperature Sensors
- Diode Temperature Coefficient
- PTAT Sensor

3.5.2 Rectifier Circuits
- Half-Wave Rectifier Circuits
- Full-Wave Rectifier Circuits
- Full-Wave Bridge Rectifier Circuits

3.5.3 Voltage Regulators

3.5.4 Additional Applications

Jaeger & Blalock, Chapter 3.13-3.16, page 113-128

3.5.1 Temperature Sensors
– Diode Temperature Coefficient –

- Forward-biased diode (operated with constant bias current) is frequently used for **temperature sensing**:

$$I_D = I_S\left[e^{qV_D/kT} - 1\right] \quad \Rightarrow \quad \boxed{V_D = \frac{kT}{q}\ln\left[\frac{I_D}{I_S} + 1\right] \cong \frac{kT}{q}\ln\left[\frac{I_D}{I_S(T)}\right]}$$

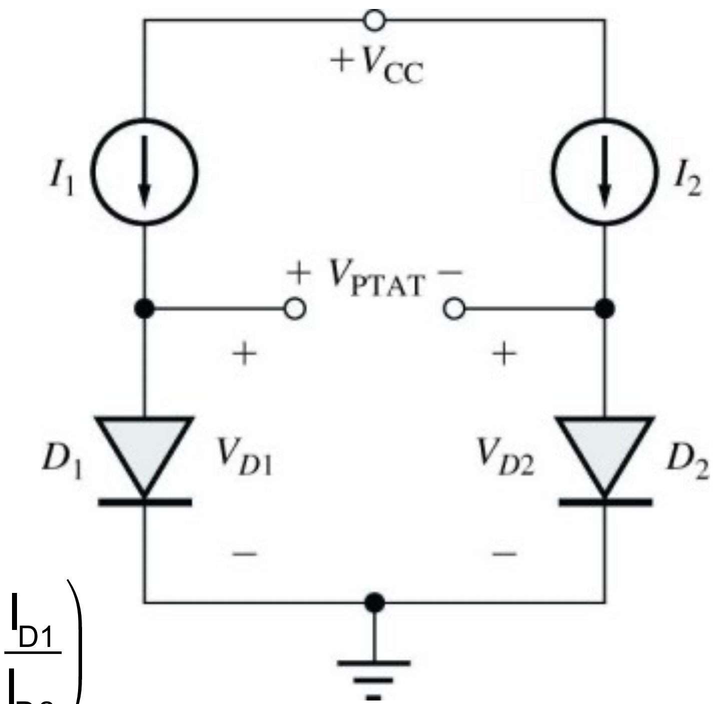
- Challenge: I_S is proportional to n_i^2 and thus (strongly) depends on temperature; as a result, the V_D vs. T relationship is nonlinear

$$\frac{dV_D}{dT} = \frac{k}{q}\ln\left[\frac{I_D}{I_S}\right] - \frac{kT}{q}\frac{1}{I_S}\frac{dI_S}{dT}$$

- Work-around: PTAT circuit using a differential setup with 2 identical diodes

PTAT Electronic Thermometer

- PTAT = **proportional to absolute temperature**

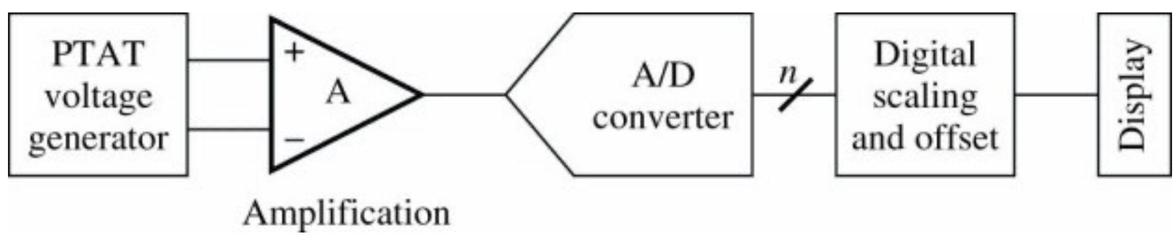


Concept

- Two identical diodes biased by current sources I_1 and I_2
- Resulting PTAT voltage, i.e., difference in voltage drop across the two diodes is proportional to the absolute temperature

$$V_{PTAT} = V_{D1} - V_{D2} = \frac{kT}{q}\left[\ln\left(\frac{I_{D1}}{I_S}\right) - \ln\left(\frac{I_{D2}}{I_S}\right)\right] = \frac{kT}{q}\ln\left(\frac{I_{D1}}{I_{D2}}\right)$$

$$\frac{dV_{PTAT}}{dT} = \frac{k}{q}\ln\left(\frac{I_{D1}}{I_{D2}}\right) = \frac{V_{PTAT}}{T}$$

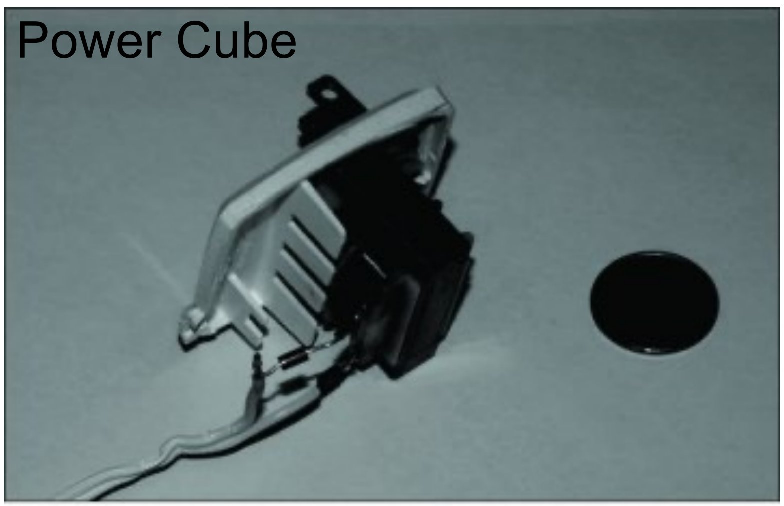


- PTAT voltage circuit is heart of most of today's **digital thermometers**

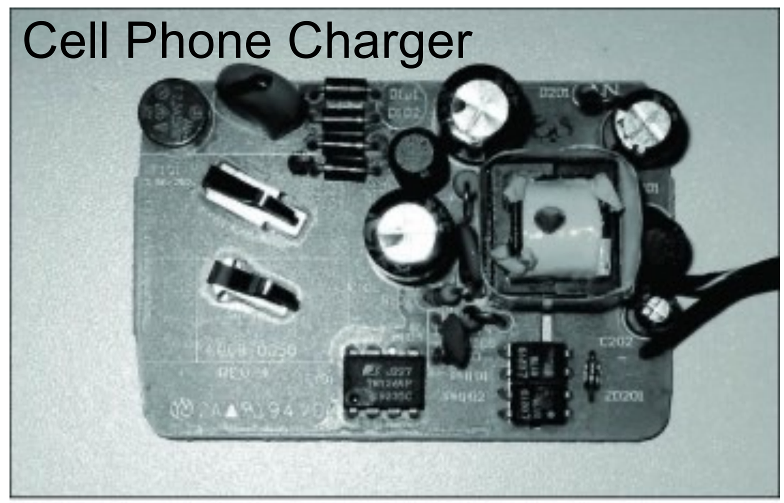
Jaeger, Blalock, page 88

3.5.2 Rectifier Circuits

- Rectifier circuit **converts an ac voltage to a** pulsating **dc voltage**; in combination with a filter, a nearly constant dc output voltage can be generated
- Virtually every electronic device plugged into the wall utilizes a rectifier circuit to convert the 120-240V, 50-60Hz ac power line source to a proper dc voltage
- DC power supplies are a commodity and fairly inexpensive



(a)

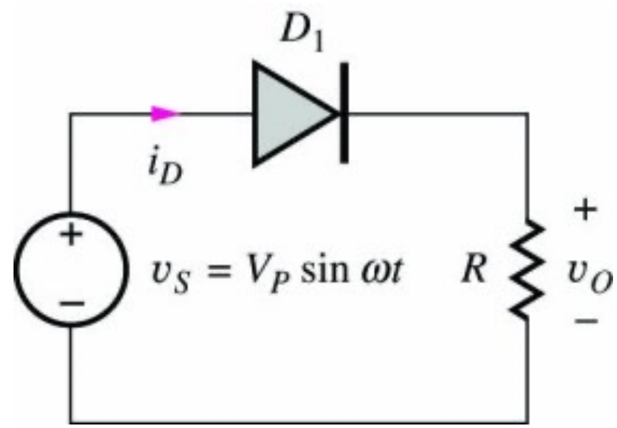


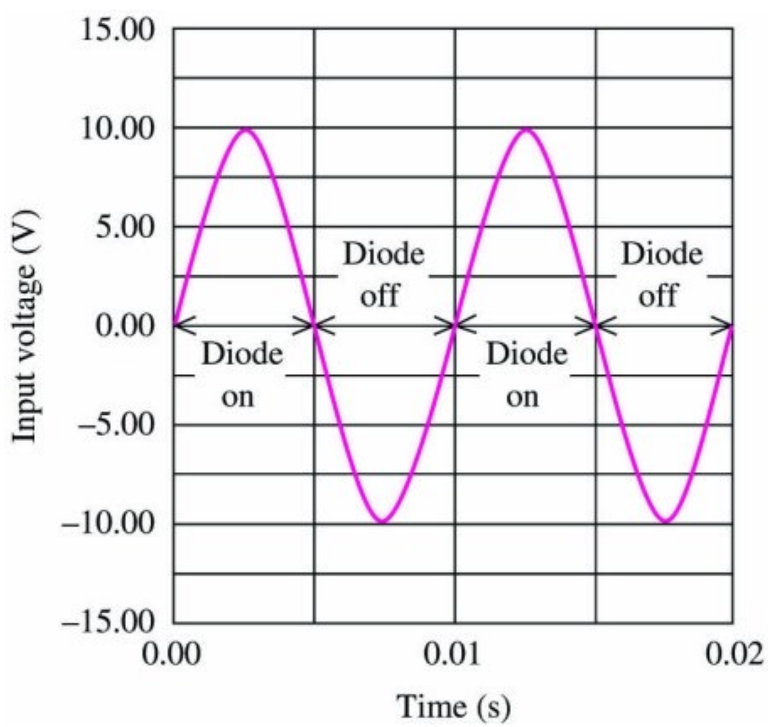
(b)

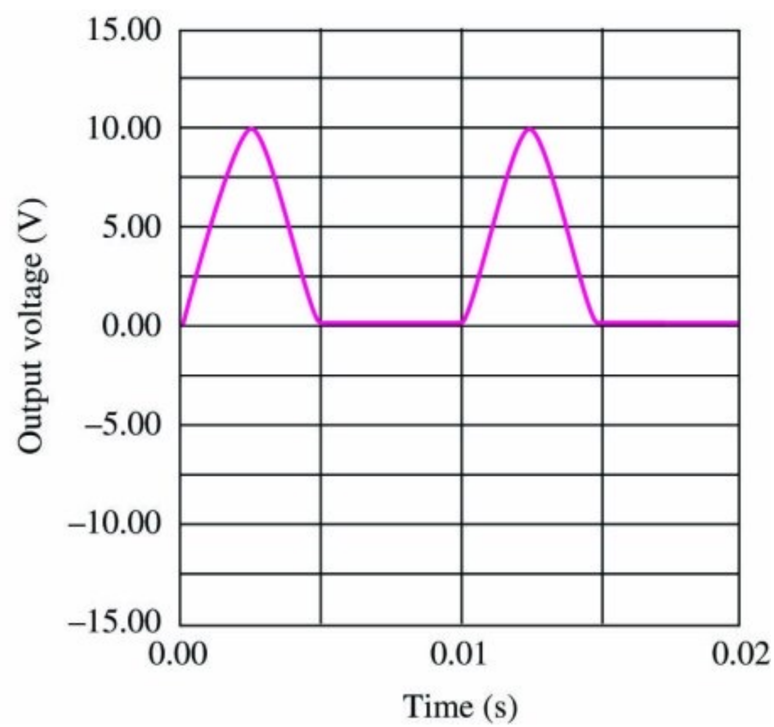
Jaeger, Blalock, page 128

Half-Wave Rectifier with Resistive Load

- Sinusoidal voltage source $v_S = V_P \sin\omega t$ is connected to a series combination of diode D_1 and resistor R
- Using the ideal diode model, we find that the diode is ON for $v_S > 0$ and OFF for $v_S < 0$, resulting in a **pulsating output voltage v_0**







Jaeger, Blalock, Fig. 3.42 & 3.44

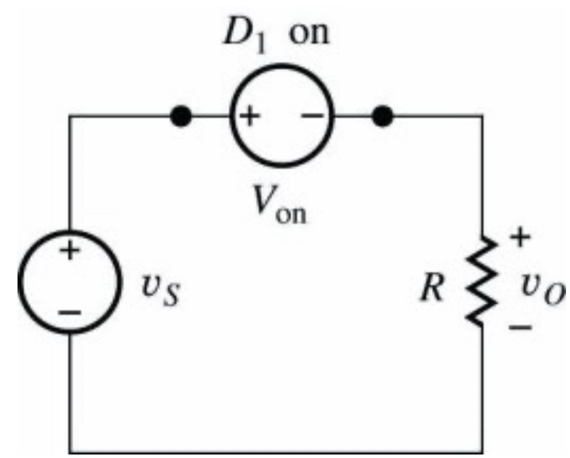
Half-Wave Rectifier with Resistive Load

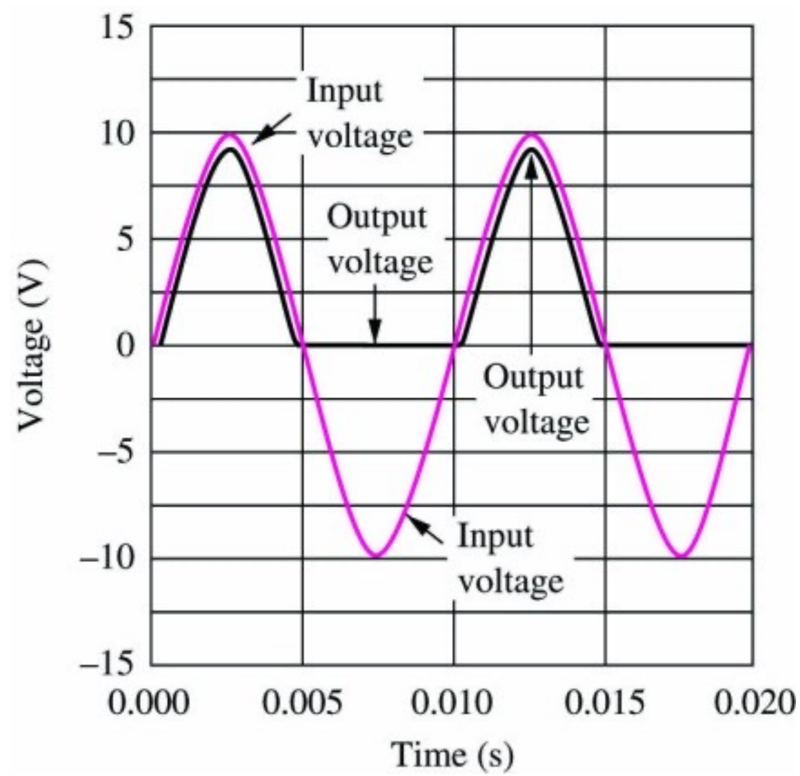
- If the input voltage amplitude (e.g. $V_P \leq 10$ V) is not large compared to the voltage drop (0.5-1.0 V) across the forward-biased diode, the CVD model should be used for circuit analysis

$$v_S \geq 0 \quad v_0 = V_P \sin \omega t - V_{on}$$

$$v_S < 0 \quad v_0 = 0$$

- In many applications, a **transformer** is used to step-down the power line voltage to a desired level
- To remove time-varying components from output waveform, a **filter capacitor** is usually added



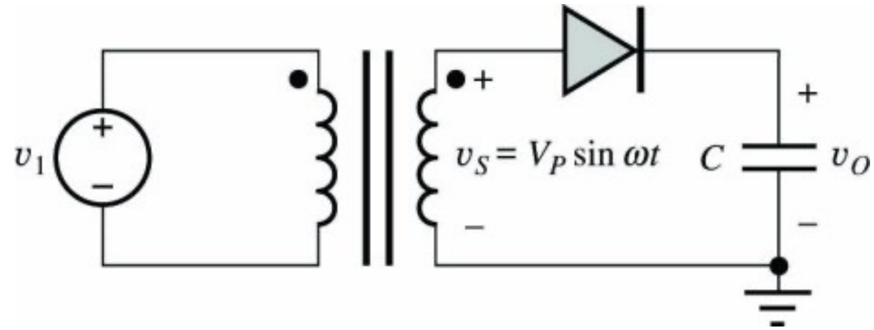


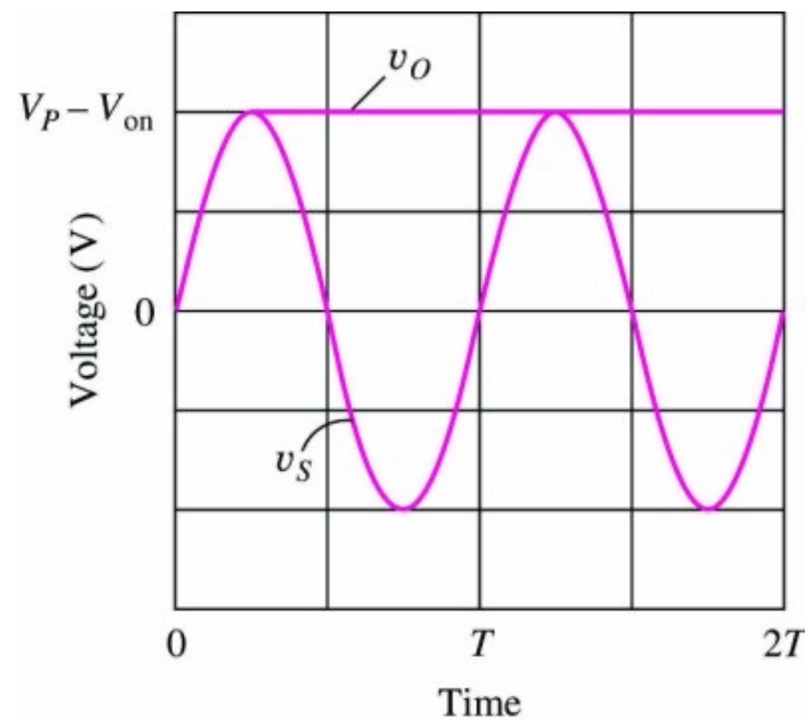
Jaeger, Blalock, Fig. 3.45 & 3.46

Rectifier Filter Capacitor

Peak Detector Circuit

- Assumption: initially uncharged capacitor, i.e. $v_0(0)=0$
- As input voltage rises, diode turns on and capacitor is charged to $v_0 = V_{dc} = V_P – V_{on}$
- At peak of input voltage waveform, the current through the diode tries to reverse direction (because $i_D = C\, d(v_S – V_{on})/dt$), thus reverse biasing the diode and disconnecting the capacitor from the power supply
- With no discharge possibility, $v_0 = V_{dc} = V_P – V_{on}$ stays constant; in a rectifier however, a resistive load R is connected in parallel to C, providing a discharge path

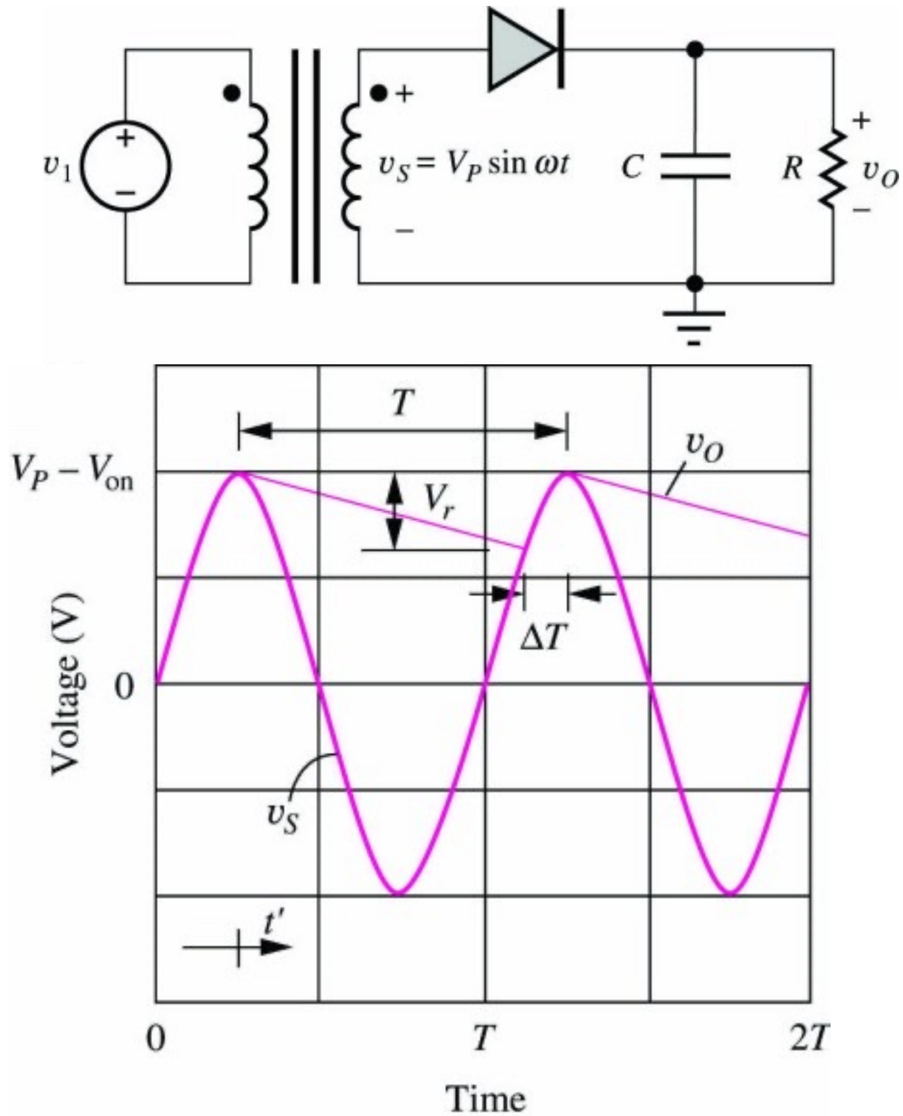




Jaeger, Blalock, Fig. 3.48 & 3.50

Half-Wave Rectifier with RC Load

- **$v_0 > v_S - V_{on}$**: diode off; capacitor discharged through resistor R
- **$v_0 < v_S - V_{on}$**: diode on; capacitor charged by voltage source
- During discharge process, the voltage across the capacitor decreases exponentially with a **time constant RC**; slow voltage decay requires large RC and thus large capacitances C
- Characteristics of rectifier circuit
  - Output voltage V_{dc}
  - Ripple voltage V_r
  - Conduction interval ΔT
  - Peak diode current I_P
  - Surge current I_{SC}
  - Peak-Inverse-Voltage Rating
  - Diode power dissipation



Jaeger, Blalock, Fig. 3.51 & 3.53

Ripple Voltage V_r

- The ripple voltage V_r should be as small as possible
- Voltage across capacitor during discharge process

$$v_0(t') = \left(V_P - V_{on}\right) e^{-t'/RC} \quad \text{for} \quad t' = t - \frac{T}{4} \geq 0$$

- The ripple voltage is thus given by

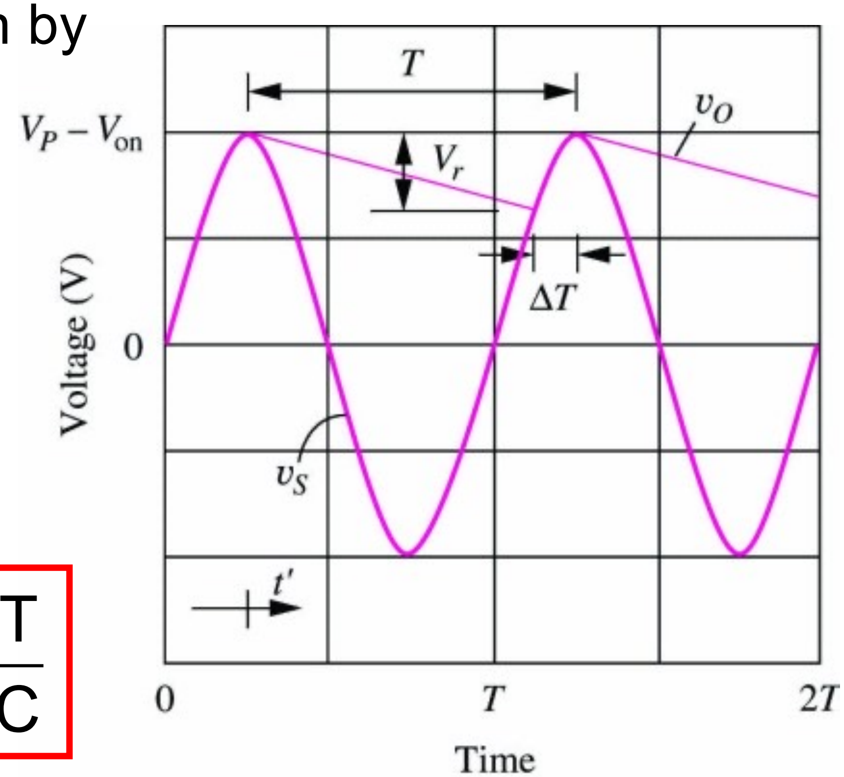
$$V_r = v_0\left(t'=0\right) - v_0\left(t'=T-\Delta T\right)$$

$$= \left(V_P - V_{on}\right)\left[e^0 - e^{-\left(T-\Delta T\right)/RC}\right]$$

$$\overset{T-\Delta T \ll RC}{\cong} \left(V_P - V_{on}\right)\left[1 - \left(1 + \frac{T-\Delta T}{RC}\right)\right]$$

$$\overset{\Delta T \ll T}{\cong} \underbrace{\frac{\left(V_P - V_{on}\right)}{R}}_{=I_{dc}} \frac{T}{C}$$

$$\boxed{V_r = I_{dc}\frac{T}{C}}$$



Conduction Interval ΔT

- At the "beginning" of the conduction interval t' = T–ΔT, we have

$$v_0(t' = T - \Delta T) = v_s(t' = T - \Delta T) - V_{on}$$
$$\left(V_P - V_{on}\right) e^{-(T-\Delta T)/RC} = V_P \cos\left[\omega\left(T - \Delta T\right)\right] - V_{on}$$

- Assuming T«RC and ΔT«T, both cosine and exponential function can be expanded in a Taylor series (cos(T-ΔT) = cos(ΔT))

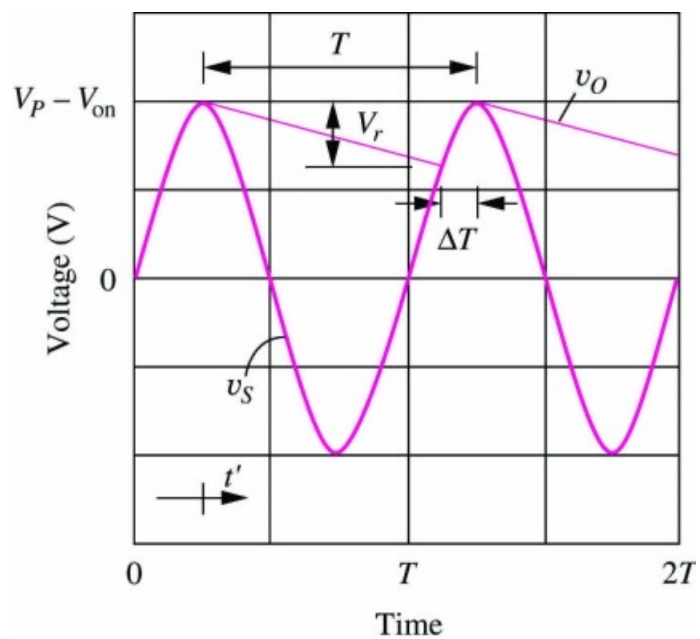
$$\left(V_P - V_{on}\right)\left[1 - \frac{T - \cancel{\Delta T}}{RC}\right] = V_P\left[1 - \frac{\left(\omega \Delta T\right)^2}{2}\right] - V_{on}$$

- Solving for ΔT yields

$$\boxed{\Delta T \cong \frac{1}{\omega}\sqrt{\frac{2T}{RC}\frac{V_P - V_{on}}{V_P}} = \frac{1}{\omega}\sqrt{\frac{2V_r}{V_P}}}$$

- Conduction angle θ_C

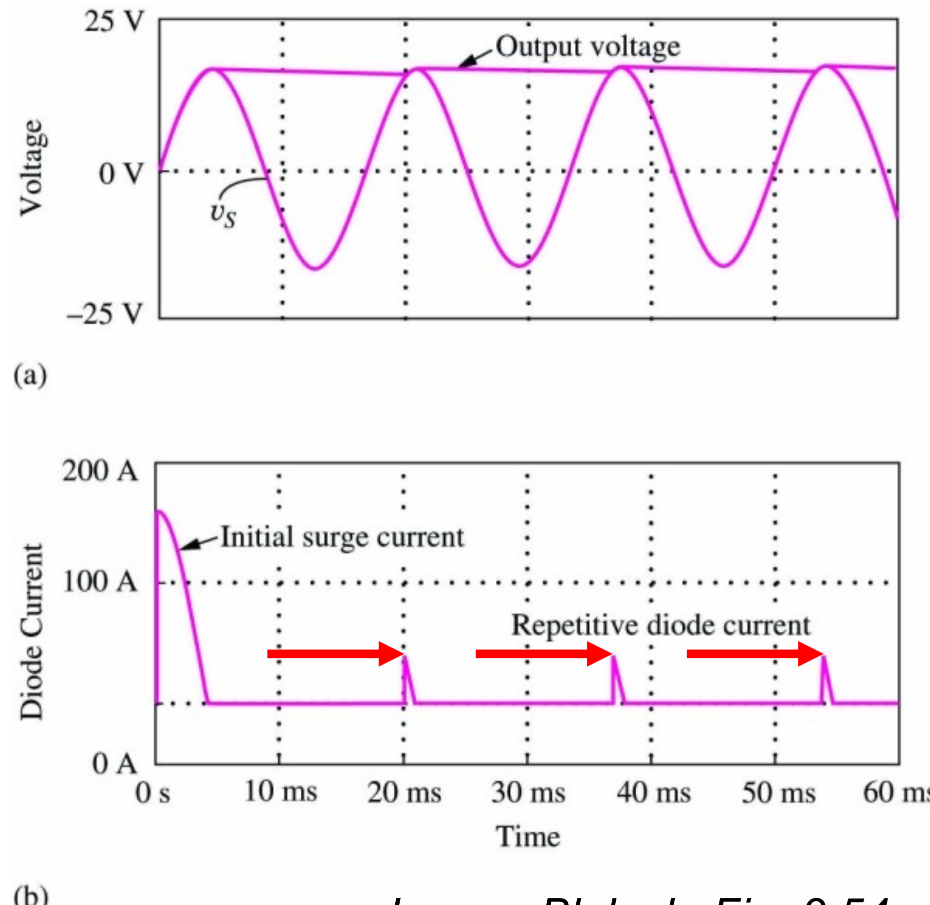
$$\theta_C = \omega\,\Delta T \cong \sqrt{\frac{2V_r}{V_P}}$$



Peak Diode Current I_P

- The charge lost during the discharge period must be supplied to the capacitor during the short charging cycle, resulting in large diode currents
- Assuming $\Delta T \ll T$, the charge "lost" during one period T is $Q = I_{dc} T$; this charge must be supplied to the capacitor during the charging interval ΔT
- Assuming a triangular shaped current pulse through the diode (see SPICE simulation), the resulting peak diode current I_P becomes

$$Q = I_{dc} T = I_P \frac{\Delta T}{2} \quad \Rightarrow \quad \boxed{I_P = I_{dc} \frac{2T}{\Delta T}}$$



Jaeger, Blalock, Fig. 3.54

Note: the peak current can be 10s of ampere

Surge Current I_{SC}

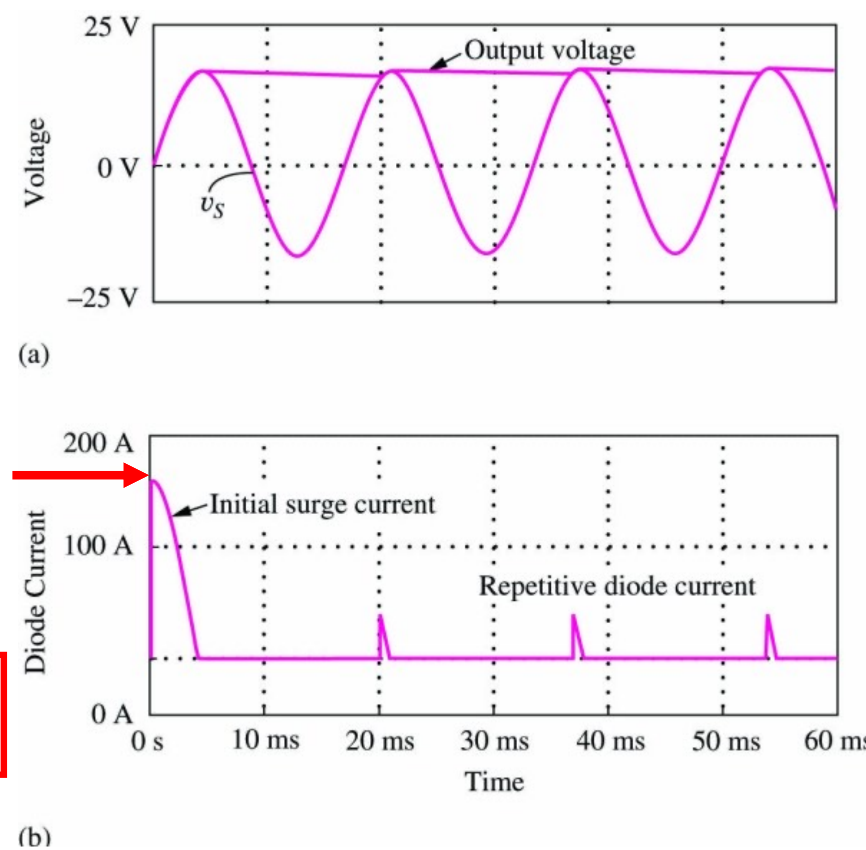
- Surge current I_{SC} = Peak current during initial charging process of capacitor
- During the initial charging, the diode current is determined by the charging current of the (uncharged) capacitor

$$i_d(t) = i_C(t) \cong C\left[\frac{d}{dt} V_P \sin\omega t\right]$$

$$= \underbrace{\omega C V_P}_{= I_{SC}} \cos\omega t$$

$$\boxed{I_{SC} = \omega C V_P}$$

- In actual devices, I_{SC} is reduced due to series resistances of diode and transformer (you can investigate this with SPICE)



Jaeger, Blalock, Fig. 3.54

Peak-Inverse-Voltage (PIV) Rating and Diode Power Dissipation

- **Peak-Inverse-Voltage (PIV)** is maximum reverse bias voltage of rectifier diode during operation; thus, minimal breakdown voltage must be PIV (typically PIV is given with safety margin of 25-50%)

$$PIV \geq V_{dc} - v_S^{min} = 2V_P - V_{on} \cong 2V_P$$

- **Power dissipation** in rectifier diode

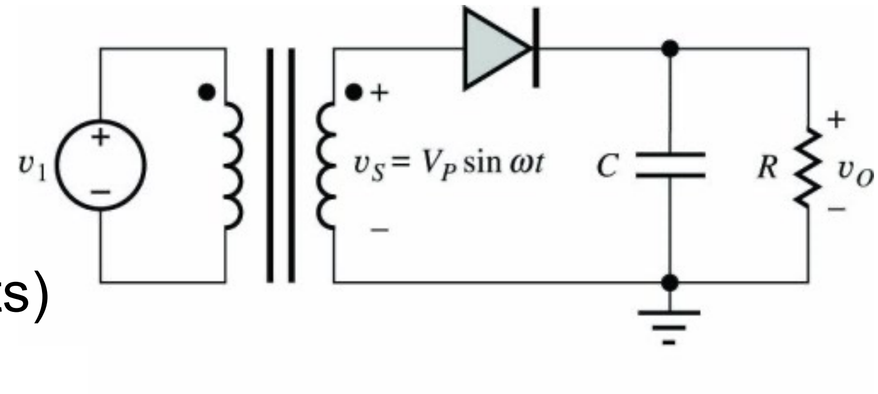
$$P_{D1} = \frac{1}{T}\int_0^T v_D(t)\, i_D(t)\, dt = \frac{V_{on}}{T} \underbrace{\int_{T-\Delta T}^{T} i_D(t)\, dt}_{\text{current only flows during charging cycle}} = V_{on}\frac{I_P}{2}\frac{\Delta T}{T} = V_{on} I_{dc}$$

Additional power dissipation due to diode series resistance R_S

$$P_{D2} = \frac{1}{T}\int_0^T R_S\, i_D^2(t)\, dt = \frac{R_S}{T}\int_{T-\Delta T}^{T} i_D^2(t)\, dt = \frac{I_P^2 R_S}{3}\frac{\Delta T}{T} = \frac{4}{3}\frac{T}{\Delta T} R_S I_{dc}^2$$

Note: P_{D2} can be substantially larger than P_{D1}

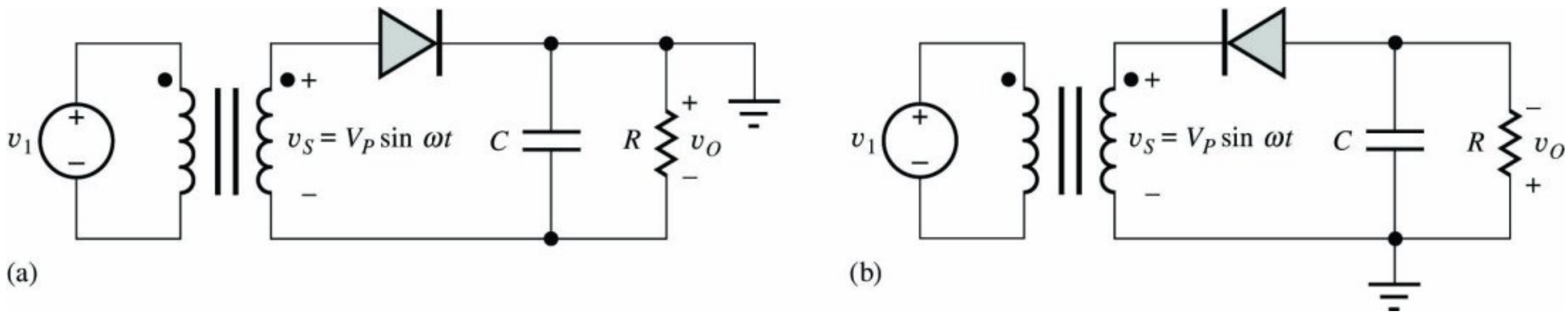
Half-Wave Rectifier Example



- Assume:
  V_P = 10 V_{rms} = 14.1 V_p (60 Hz)
  R = 15 Ω
  C = 25,000 μF !!
  V_{on} = 1 V (because of large currents)
  R_S = 0.2 Ω

- DC output voltage V_{dc}: $V_{dc} = V_P - V_{on} = 13.1\text{V}$
- Output current I_{dc}: $I_{dc} = V_{dc} / R = 0.87\,\text{A}$
- Ripple voltage V_r: $V_r = (I_{dc}\,T) / C = 0.58\text{ V}$
- Conduction interval ΔT and angle θ_C: $\Delta T \cong \omega^{-1}\sqrt{2V_r / V_P} = 0.761\,\text{ms}$
  $\theta_C = \omega\,\Delta T = 0.287\text{ rad} = 16.4^\circ$
- Diode peak current I_P: $I_P = (2\,I_{dc} T) / \Delta T = 38\text{ A}$
- Diode surge current I_{SC}: $I_{SC} = \omega\,C\,V_P = 133\text{ A}$
- PIV: $\text{PIV} \geq 2V_P = 28.2\,\text{V}$
- Diode power dissipation: $P_{D1} = V_{on} I_{dc} = 0.87\,\text{W}$ $P_{D2} = \frac{4}{3}\frac{T}{\Delta T} R_S I_{dc}^2 = 4.42\text{ W}$

Half-Wave Rectifier with Negative Output Voltage



Jaeger, Blalock, Fig. 3.57

- By grounding either the top of the capacitor (compared to the bottom in Jaeger/Blalock, Fig. 3.48) or reversing the diode, the rectifier circuit produces a **negative output voltage**, i.e. the diode conducts on the negative half-cycle of the transformer

$$V_{dc} = -\left(V_P - V_{on}\right)$$

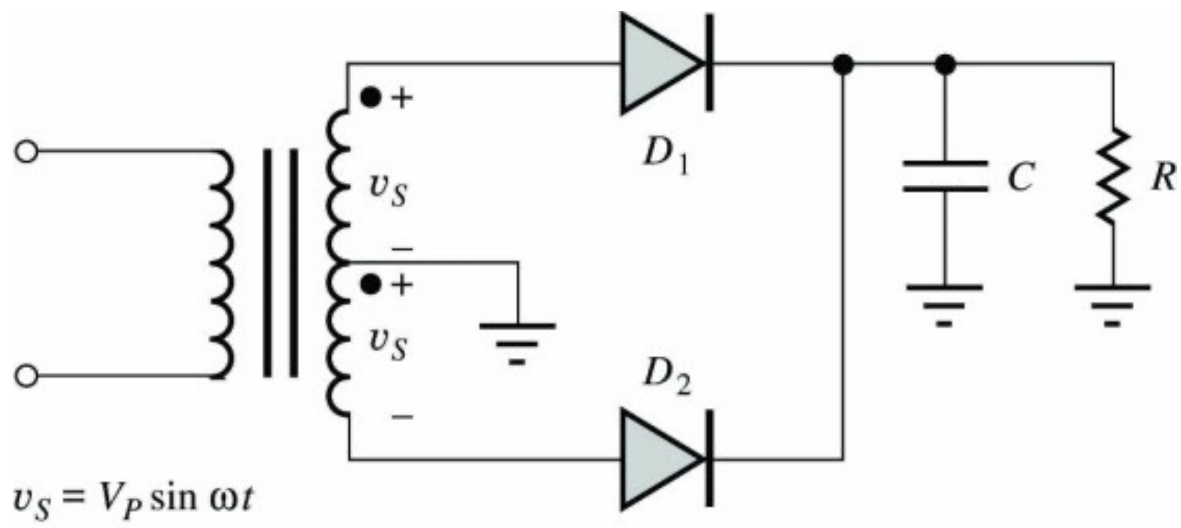
Full-Wave Rectifier Circuits

Operation Principle

- **Center-tapped transformer** generates two voltages with equal amplitude but 180˚ phase shift
- Resulting are **two half-wave rectifier circuits operating on alternate half-cycles** of input waveform
- D_1 charges C during one half cycle, D_2 charges C during the other half cycle

Advantages

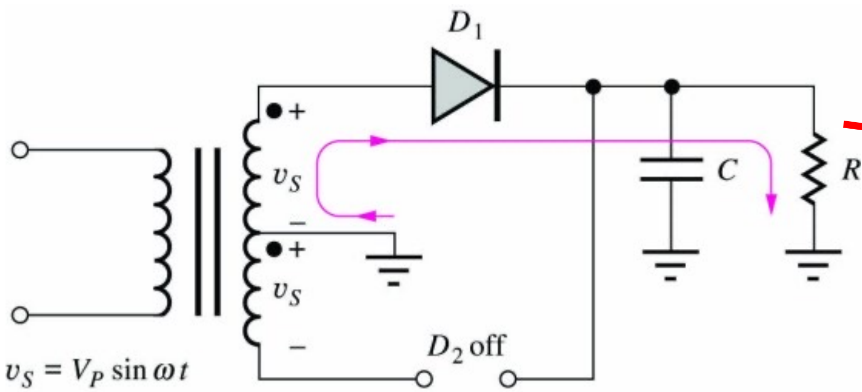
- Capacitor discharge time cut in half
- Thus, requires only one-half the filter capacitance for a given ripple voltage



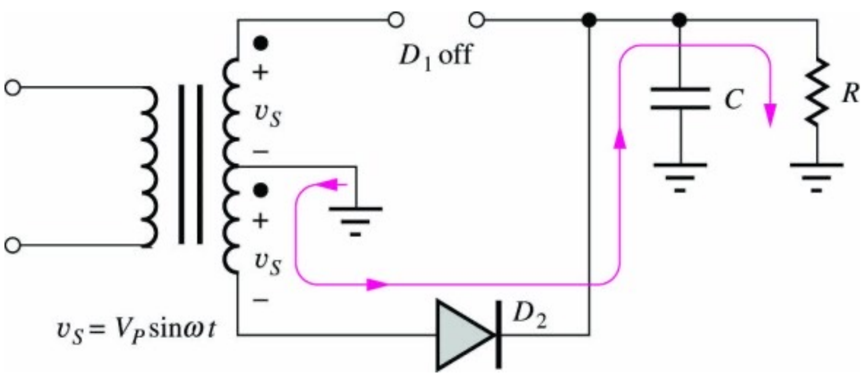
Jaeger, Blalock, Fig. 3.58

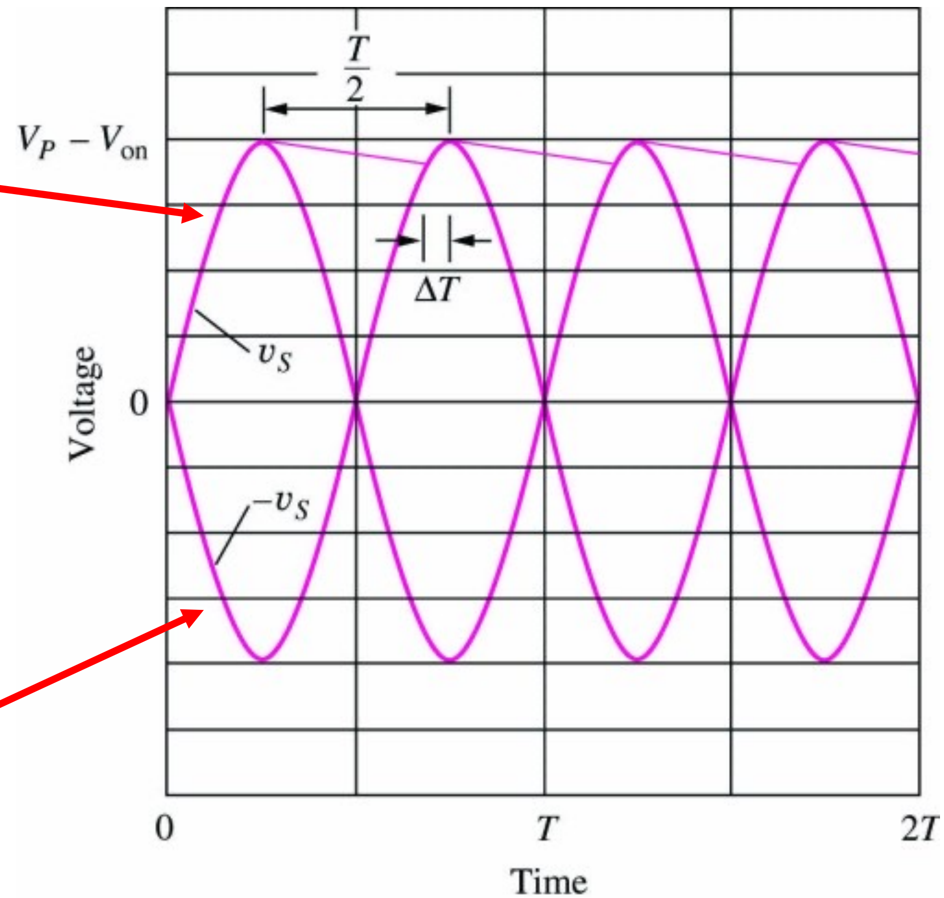
Full-Wave Rectifier Circuit

- Equivalent circuit for $v_S > 0$:



- Equivalent circuit for $v_S < 0$:





Jaeger, Blalock, Figs. 3.59-3.61

RC load receives 2 current pulses per cycle!

Full-Wave Rectifier Circuit

- DC output Voltage V_{dc}:
- Ripple Voltage V_r:
- Conduction Interval ΔT:
- Conduction Angle θ_C:
- Peak Diode Current I_P:
- PIV:

$$V_{dc} = \left(V_P - V_{on}\right)$$

$$V_r = \frac{\left(V_P - V_{on}\right)}{R}\frac{T}{2C} = I_{dc}\frac{T}{2C}$$

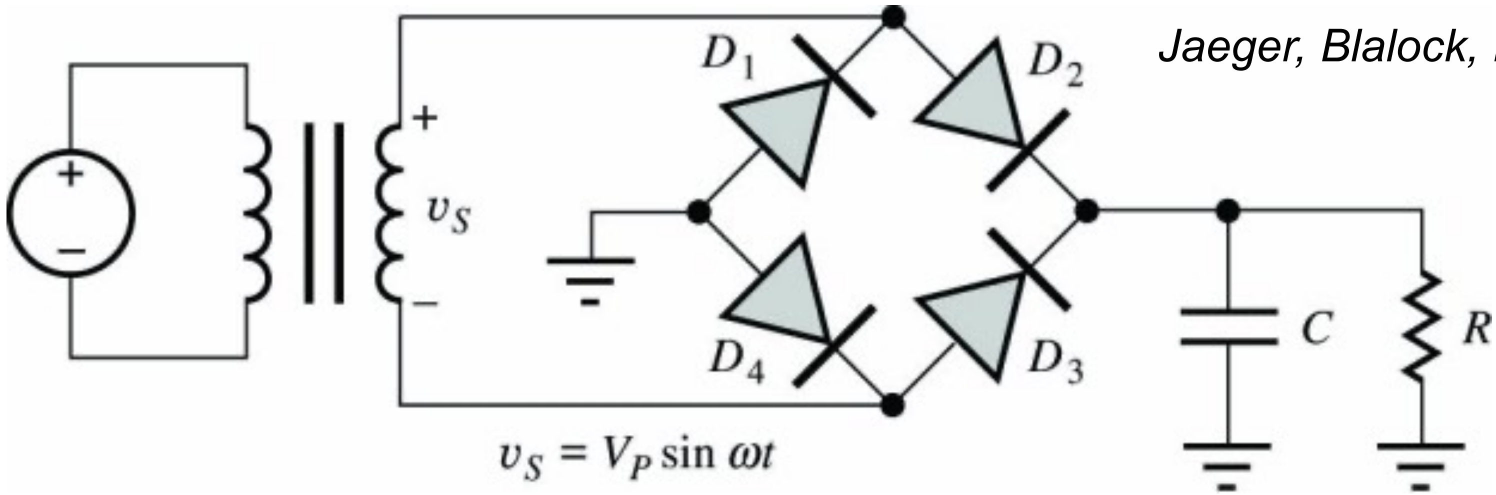
$$\Delta T \cong \frac{1}{\omega}\sqrt{\frac{2T}{2RC}\frac{V_P - V_{on}}{V_P}} = \frac{1}{\omega}\sqrt{\frac{2V_r}{V_P}}$$

$$\theta_C = \omega\Delta T \cong \sqrt{\frac{2V_r}{V_P}}$$

$$I_P = I_{dc}\frac{2T}{2\Delta T}$$

$$PIV = 2\,V_P$$

Full-Wave Bridge Rectification



Jaeger, Blalock, Fig. 3.63

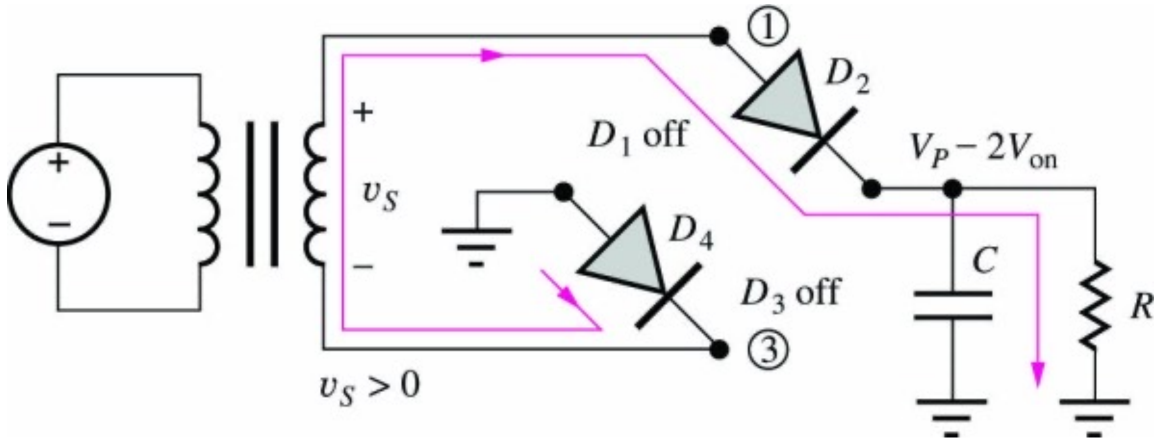
- Bridge arrangement of 4 diodes eliminates the need for a center-tapped transformer
- $v_S > 0$ half cycle: D_2 and D_4 ON; D_1 and D_3 OFF
- $v_S < 0$ half cycle: D_1 and D_3 ON; D_2 and D_4 OFF
- Output dc voltage is now reduced by 2 diode voltage drops

$$\boxed{V_{dc} = V_P - 2V_{on}}$$

- PIV rating for each diode is reduced: PIV ≈ V_P

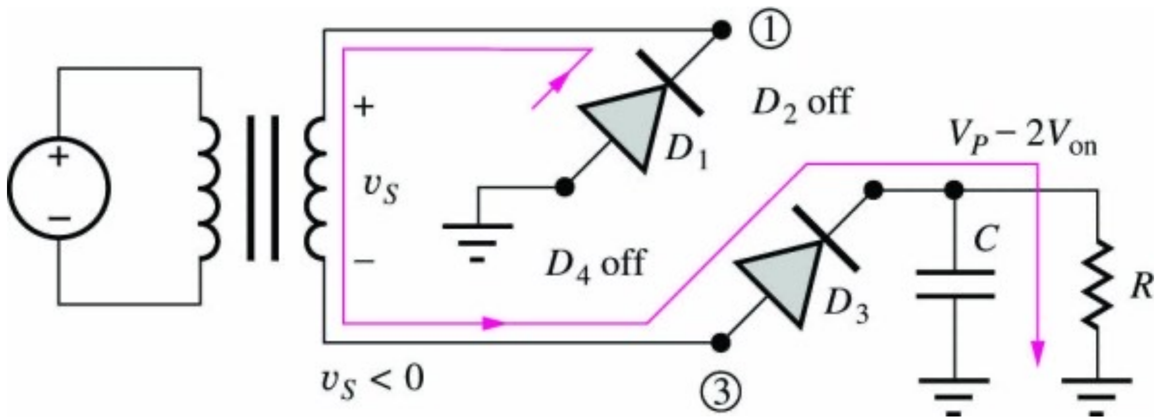
Full-Wave Bridge Rectification

Half Cycle $v_S > 0$:



Half Cycle $v_S < 0$:

Jaeger, Blalock, Figs. 3.64 and 3.65

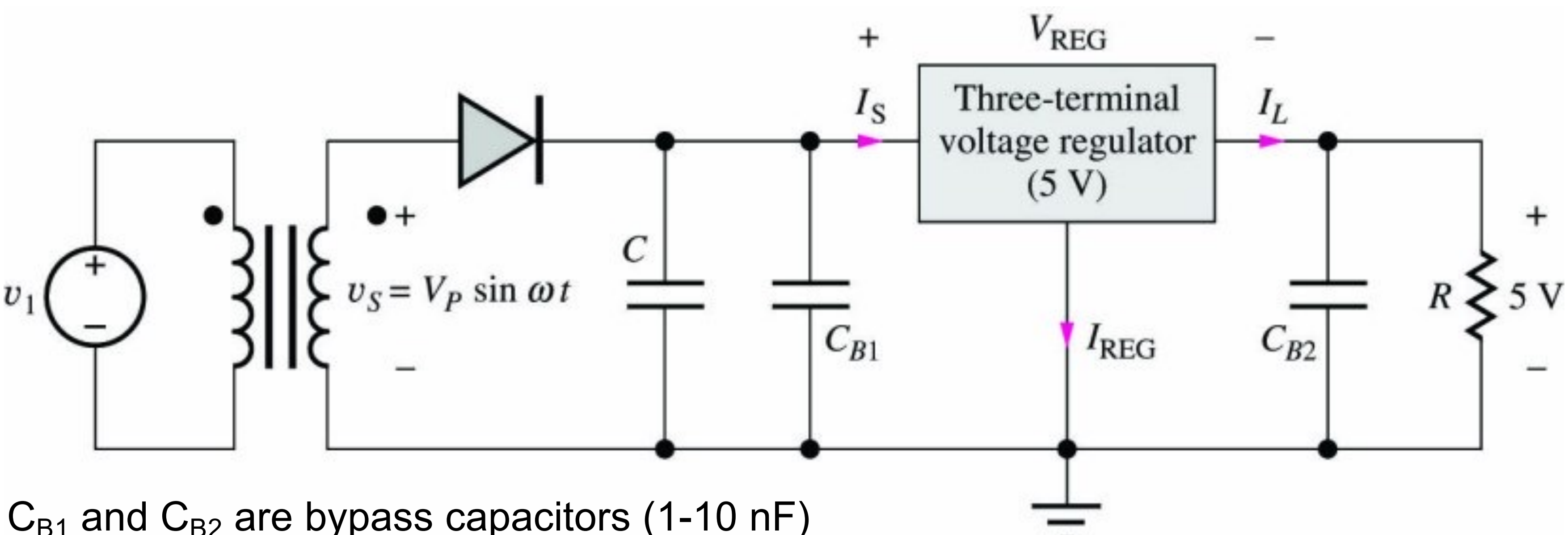


Rectifier Comparison

Rectifier Parameter	Half-Wave Rectifier	Full-Wave Rectifier	Full-Wave Bridge Rectifier
Filter Capacitor	$C = \frac{V_P - V_{on}}{V_r}\frac{T}{R}$	$C = \frac{V_P - V_{on}}{V_r}\frac{T}{2R}$	$C = \frac{V_P - V_{on}}{V_r}\frac{T}{2R}$
PIV Rating	$2\ V_P$	$2\ V_P$	V_P
Peak Diode Cur. (const. V_r)	Highest I_P	Reduced $I_P/2$	Reduced $I_P/2$
Comments	Least complexity	Smaller capacitor Requires center-tapped transformer 2 diodes	Smaller capacitor 4 diodes

3.5.3 IC Voltage Regulator

- Precise dc output voltages (under changing loads) generated by rectifier circuits are challenging, requiring transformers with specific windings and ultra-large capacitor filters to minimize V_r
- Alternative, integrated circuit voltage regulators can be used to set V_{dc} and remove the voltage ripples of a half-wave rectifier (see below)
- Via feedback with a high-gain amplifier, the regulator maintains a constant output voltage, reducing V_r by a factor of 10^2-10^3 or more



C_{B1} and C_{B2} are bypass capacitors (1-10 nF)

Jaeger, Blalock, 3rd ed., page 126

3.5.4 Additional Diode Applications
(not discussed in ECE3040)

- DC-to-DC Converters
- Wave-Shaping Circuits
  - Clamping or DC Restoring Circuit
  - Clipping or Limiting Circuit
  - Piecewise Linear Transfer Function Circuit
- Optoelectronic Devices
  - Photo Diodes
  - Solar Cells
  - Light-Emitting Diodes